3D Unstructured Spectral Element Method for Frequency-Domain Airborne EM Forward Modeling Based on Coulomb Gauge

Jiao Zhu, Changchun Yin, Lingqi Gao, Zhejian Hui, Yunhe Liu, Xiuyan Ren, Bo Zhang, Junbo Wang, and Bin Xiong

***Abstract*— In this article, to solve the frequency-domain airborne electromagnetic (EM) modeling problem, we express the electric and magnetic field by the vector magnetic potential and the scalar electric potential and derive the governing equation using the Galerkin weighted residual method under the Coulomb gauge. To avoid the strong singularity of the solution near the transmitting source, we directly solve the relatively slow-changing secondary potential. By introducing the spectral element method (SEM) based on unstructured tetrahedral grids, in which the EM field in each element is characterized by high-order Proriol–Koornwinder–Dubiner (PKD) orthogonal polynomials, we can obtain stable and accurate numerical solutions. By establishing the mapping relationship between the physical domain, the right-angled tetrahedral reference domain, and the orthogonal hexahedral domain, we can easily accomplish the element matrix analysis and calculation. The numerical examples show that, when the SEM is used to simulate the airborne EM response, a high accuracy can be obtained even if a very coarse grid is used. Furthermore, since the tetrahedral grids can easily model the complex boundaries, the SEM based on unstructured grids has significant advantages for simulating complex underground structures.**

***Index Terms*— Airborne electromagnetic (AEM), Coulomb gauge, forward modeling, spectral element method (SEM), unstructured grid.**

I. Introduction

In RECENT years, with the increasing demand for mineral resources, the focus of geophysical exploration begins to shift to areas with intricate working conditions. The airborne electromagnetic (AEM) system carries the transmitting and receiving devices on aircraft that is free from the restriction of the rugged topography encountered by the traditional ground EM, and its efficiency is high, while the cost is low. Thus, the AEM method is very suitable for exploration in areas where personnel access is difficult, such as mountains, deserts, forest-covered areas, lakes, and swamps [1].

With the continuous advancement of AEM technology, challenges have been raised to forward modeling and inversions. The existing EM forward modeling methods, such as the integral equation (IE), the finite difference (FD), finite volume (FV), and finite-element (FE) method, all rely heavily on grids. Among them, the IE method uses hexahedral grids [2], while the FD and FV mostly use Yee's staggered hexahedral grids, so the interfaces in the underground are frequently discretized in a zigzag shape, resulting in step effect [3], [4]. Although the orthogonal Delaunay–Voronoï unstructured grids can be used in the FV method, only its implementation is very complicated [5]. Mathematically, the FE method can easily simulate regular and irregular grids. Except for the commonly used hexahedral and tetrahedral grids, the prism or pyramid grids have also been used for FE [6]. Therefore, the FE method has the flexibility in modeling complex structures.

In the early time, the nodal FE method was used to solve EM problems, but it encountered the following problems: 1) the discontinuity occurs at the interfaces of different media and 2) the undefined divergence can lead to the nonphysical solutions or spurious modes, while the divergence correction cannot completely eliminate the spurious modes. This deteriorates the calculation accuracy [7]. Nédéléc [8] introduced the vector element that not only ensured the continuity of the field at interfaces but also automatically satisfied the condition of vanishing divergence within each element so that the calculation accuracy was greatly improved. In addition to calculating the field directly, scholars have also tried to indirectly solve EM problems by introducing potentials. At present, the commonly used strategies include $\mathbf{A}-\Phi$, $\mathbf{T}-\Omega$, and $\mathbf{A}-\Phi-\Psi$ methods and so on. [9]. Among them, the $\mathbf{A}-\Phi$ method is widely used due to its simplicity in implementation. In EM geophysics, Badea *et al.* [10] elaborated on the advantages of the potential method by applying it to simulate the controlled-source EM problem; Puzyrev *et al.* [11] studied the singularity removal in EM modeling of anisotropic media; and Ansari *et al.* [12] used the potential to model EM fields

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for ovoid ore deposit in Voisey's Bay, Canada. The studies above have all demonstrated that: 1) the potential method can obtain more information of both potentials and EM fields; 2) since the tangential and normal components of $\mathbf{A}$ and Φ are continuous, there is no need to use vector elements to remove spurious modes, which can be applied to the case of multiply connected domains [9]; 3) the potential method usually has higher stability than the direct solution of the EM field, and the system matrix has better conditions [13]; 4) the nonuniqueness of the solution can be eliminated by applying the Coulomb gauge or the Lorentz gauge [14]; and 5) the potential method can take many techniques for nodal element, such as the meshless method, the spectral element method (SEM), and the wavelet FE method, to effectively do EM forward modeling [15]–[17].

Compared to other numerical methods, the FE method can obtain higher calculation accuracy. However, it needs fine grids. This sometimes brings challenges, especially when the computational resource is limited. Moreover, in practical applications, when the mesh is refined to a certain level, the convergence rate decreases gradually, while the number of grids and the computational cost increase largely [18]. Thus, in this article, we introduce the spectral method. In this method, the field in the calculation domain is approximated by polynomials (nonperiodic problems) or Fourier expansions (periodic problems), and then, we use these high-precision basis functions as test functions to discretize partial differential equations. According to the solution strategy, the spectral method can be divided into the Galerkin method, the tau method, and the collocation method (or the pseudospectral method) [19]. Since the spectral method adopts a high order of orthogonal basis functions, it has exponential convergence. In addition, the spectral interpolation points are densely distributed at the boundary, which can overcome the Runge phenomenon in the traditional high-order interpolation [20]. The method that combines the mesh flexibility of FE and the high precision of the spectral method is called the SEM. Petera [21] first proposed SEM and successfully applied it in fluid dynamics. Subsequently, Rønquist and Patera [22] used Gauss–Lobatto–Legendre (GLL) polynomials as the basis functions to solve the multidimensional Stefan problem; Seriani [23] utilized SEM to simulate the acoustic wave field in a double-grid; and Lee *et al.* [24] verified the effectiveness of SEM in solving EM waveguide problems via theoretical and numerical studies. In seismic exploration, SEM based on hexahedral grids has been widely used [25]. Komatitsch *et al.* [26]–[28] successfully released their open-source code SPECFEM3D, and they further improved the performance of the algorithm and solved the problems of perfectly matched layer, domain decomposition, and parallel calculation. The SEM code has also been used to simulate the global seismic wave field [26]–[28]. Liu *et al.* [29] used the mass lumping technique and GLL quadrature rule to study the triangular spectral elements for the modeling of the time-domain elastic wave field. In gravity exploration, Roland *et al.* [30] combined SEM based on GLL points with the least-squares method for the inversion of gravity data, and they performed the geological interpretation of the field data acquired in Mauléon sedimentary basin, France. In EM exploration, Yin *et al.* [31], Liu *et al.* [32], and Huang *et al.* [33] introduced SEM based on hexahedral grids into airborne and marine EM and achieved good modeling results.

In SEM, based on hexahedral grids, the points corresponding to the standard tensor basis functions have good interpolation properties and high calculation accuracy. However, the main drawback of the SEM using hexahedral grids lies in that it cannot effectively handle irregular anomalous bodies and sharp boundaries. Even if the deformed hexahedral grids can be used to fit the irregular surfaces, they will produce additional errors related to curvature [34]. At present, the most effective strategy is to develop the unstructured tetrahedral SEM (TSEM). This is because, on the one hand, the tetrahedral mesh can better fit the irregular surfaces, while, on the other hand, the SEM can ensure more accurate solutions. At present, TSEM is mainly used in fluid dynamics, elastic waves, computational EM, and so on [35], [36]. In EM geophysics, we combine for the first time the nodal spectral element with the $\mathbf{A}-\Phi$ method to solve the EM modeling problems. In the sequence, we will first introduce the technical details of TSEM in solving EM potentials, and then, we will verify the accuracy of our algorithm via a layered Earth model. After that, we study and analyze the impact of refining grids and increasing the order of basis functions on accuracy. Finally, we use a 3-D topographic Earth model to test the effectiveness of our TSEM for simulating complex structures.

II. Methodology

A. Governing Equations

Assuming the harmonic time factor of $e^{i\omega t}$, Maxwell's equation in an isotropic medium can be written as

$$\begin{cases} \nabla \times \mathbf{E} = -i\omega\mu_0\mathbf{H} \\ \nabla \times \mathbf{H} = \sigma\mathbf{E} + i\omega\varepsilon_0\mathbf{E} + \mathbf{J}^e \end{cases} \tag{1}$$

where $\mathbf{E}$ and $\mathbf{H}$, respectively, represent the electric and magnetic fields, $\omega = 2\pi f$ is the angular frequency, $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability in a vacuum, σ is the conductivity, and $\varepsilon_0 = (1/4\pi)\times 10^{-9}$ F/m is the dielectric constant, while $\mathbf{J}^e$ is the current density. The electric field $\mathbf{E}$ and the magnetic field $\mathbf{H}$ are expressed by the vector potential $\mathbf{A}$ and scalar potential Φ as

$$\mathbf{E} = -i\omega\mathbf{A} - \nabla\Phi \tag{2}$$

$$\mathbf{H} = \frac{1}{\mu_0}\nabla \times \mathbf{A}. \tag{3}$$

The insertion of (2) and (3) into the second equation in (1) yields

$$\nabla \times \left(\frac{1}{\mu_0}\nabla \times \mathbf{A}\right) + i\omega\sigma\mathbf{A} + \sigma\nabla\Phi = \mathbf{J}^e. \tag{4}$$

Using the equality $\nabla \times \nabla \times \mathbf{A} = -\nabla^2\mathbf{A} + \nabla(\nabla \cdot \mathbf{A})$ and multiplying μ_0 to both sides of (4), we obtain

$$-\nabla^2\mathrm{A} + \nabla(\nabla \cdot \mathbf{A}) + i\omega\mu_0\sigma\mathbf{A} + \mu_0\sigma\nabla\Phi = \mu_0\mathbf{J}^e. \tag{5}$$

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In order to eliminate the spurious modes, we apply the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$ in the solution domain and rewrite (5) as

$$-\nabla^2 A + i\omega\mu_0\sigma\mathbf{A} + \mu_0\sigma\nabla\Phi = \mu_0\mathbf{J}^e. \tag{6}$$

To ensure that all physical conditions are met, the following auxiliary condition is applied, i.e.,

$$\nabla \cdot [i\omega\mu_0\sigma\mathbf{A} + \mu_0\sigma\nabla\Phi] = 0. \tag{7}$$

According to Ansari *et al.* [12], for the inductively coupled source $\nabla\cdot\mathbf{J}^{\mathrm{e}} = 0$, however, for the directly coupled sources, it is no longer satisfied. Thus, we introduce the following general auxiliary equation, i.e.,

$$\nabla \cdot [i\omega\mu_0\sigma\mathbf{A} + \mu_0\sigma\nabla\Phi] = \nabla \cdot (\mu_0\mathbf{J}^e). \tag{8}$$

Equations (6)–(8) constitute an explicit stabilization system of Maxwell's equations based on EM potential. Aiming at AEM modeling, we use the field separation algorithm, that is,

$$\begin{cases} \mathbf{A} = \mathbf{A}_p + \mathbf{A}_s \\ \Phi = \Phi_p + \Phi_s \end{cases} \tag{9}$$

where $\mathbf{A}_p$ and Φ_p are the primary vector and scalar potentials generated by the background model, while $\mathbf{A}_s$ and Φ_s are the secondary vector and scalar potentials generated by the underground abnormal bodies. Then, the secondary potentials satisfy

$$\begin{cases} -\nabla^2\mathbf{A}_s + i\omega\mu_0\sigma\mathbf{A}_s + \mu_0\sigma\nabla\Phi_s \\ \quad = -i\omega\mu_0\sigma_s\mathbf{A}_p - \mu_0\sigma_s\nabla\Phi_p \\ \nabla \cdot [i\omega\mu_0\sigma\mathbf{A}_s + \mu_0\sigma\nabla\Phi_s] \\ \quad = -\nabla \cdot \left[i\omega\mu_0\sigma_s\mathbf{A}_p + \mu_0\sigma_s\nabla\Phi_p\right] \end{cases} \tag{10}$$

Considering that the primary field satisfies $\mathbf{E}_p = -i\omega\mathbf{A}_p - \nabla\Phi_p$, (10) can be rewritten as

$$\begin{cases} -\nabla^2\mathbf{A}_s + i\omega\mu_0\sigma\mathbf{A}_s + \mu_0\sigma\nabla\Phi_s = \mu_0\sigma_s\mathbf{E}_p \\ \nabla \cdot [i\omega\mu_0\sigma\mathbf{A}_s + \mu_0\sigma\nabla\Phi_s] = \nabla \cdot \left[\mu_0\sigma_s\mathbf{E}_p\right]. \end{cases} \tag{11}$$

In this article, we use the analytical solution for AEM system in a free space as the primary field. Thus, decomposing the vector potential $\mathbf{A}_{\mathrm{s}}$ and the electric field $\mathbf{E}_p$ in (11) into components, we have

$$\begin{cases} -\nabla^2 A_{sx} + i\omega\mu_0\sigma A_{sx} + \mu_0\sigma\dfrac{\partial\Phi_s}{\partial x} = \mu_0\sigma_s E_{px} \\ -\nabla^2 A_{sy} + i\omega\mu_0\sigma A_{sy} + \mu_0\sigma\dfrac{\partial\Phi_s}{\partial y} = \mu_0\sigma_s E_{py} \\ -\nabla^2 A_{sz} + i\omega\mu_0\sigma A_{sz} + \mu_0\sigma\dfrac{\partial\Phi_s}{\partial z} = \mu_0\sigma_s E_{pz} \\ i\omega\mu_0\sigma\nabla \cdot \mathbf{A}_s + \nabla \cdot (\mu_0\sigma\nabla\Phi_s) = \mu_0\sigma_s\nabla \cdot \mathbf{E}_p. \end{cases} \tag{12}$$

B. High-Order Basis Functions for Tetrahedral Grids

Using S^3 to represent a 3-D simplex, the dimension of n-order polynomial space can be expressed as

$$\dim N^n(S^3) = Nt = \binom{n+3}{n} = \frac{(n+1)(n+2)(n+3)}{6}. \tag{13}$$

The basis functions $N_i (i = 1, \ldots, Nt)$ for our TSEM can be defined as

$$\begin{aligned} \mathbf{N} &= \begin{bmatrix} N_1(\mathbf{r}) \\ N_2(\mathbf{r}) \\ \vdots \\ N_{Nt}(\mathbf{r}) \end{bmatrix} \\ &= \begin{bmatrix} \psi_1(\mathbf{r}_1) & \psi_2(\mathbf{r}_1) & \cdots & \psi_{Nt}(\mathbf{r}_1) \\ \psi_1(\mathbf{r}_2) & \psi_2(\mathbf{r}_2) & \cdots & \psi_{Nt}(\mathbf{r}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_1(\mathbf{r}_{Nt}) & \psi_2(\mathbf{r}_{Nt}) & \cdots & \psi_{Nt}(\mathbf{r}_{Nt}) \end{bmatrix}^{-T} \begin{bmatrix} \psi_1(\mathbf{r}) \\ \psi_2(\mathbf{r}) \\ \vdots \\ \psi_{Nt}(\mathbf{r}) \end{bmatrix} \\ &= (\mathbf{V}^{-1})^T\boldsymbol{\psi} \end{aligned} \tag{14}$$

where $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_{Nt}$ are the coordinates of interpolation nodes in the element and $\mathbf{V}$ is the Vandermonde matrix, while the element $\psi_j(\mathbf{r}_i)$ is the Proriol–Koornwinder–Dubiner (PKD) orthogonal polynomial, defined by

$$\begin{aligned} &\psi_m(\xi, \eta, \zeta) \\ &\quad = \sqrt{\frac{8}{\gamma_{ijk}}} P_i^{(0,0)}\left(-2\frac{1+\xi}{\eta+\zeta} - 1\right) \\ &\qquad \times P_j^{(2i+1,0)}\left(2\frac{1+\eta}{1-\zeta} - 1\right)\left(2 - 2\frac{1+\eta}{1-\zeta}\right)^i \\ &\qquad \times P_k^{(2i+2j+2,0)}(\zeta)(1-\zeta)^{i+j} \end{aligned} \tag{15}$$

where $\gamma_{ijk} = (2/2i+1)(2^{2i+2}/2i+2j+2)(2^{2i+2j+3}/2i+2j+2k+3)$ are orthogonal coefficients, satisfying $i, j, k \geq 0$, $i + j + k \leq n$, and m is a composite index related to n. More exactly, we have $\sum_{m=1}^{Nt} \Leftrightarrow \sum_{i=0}^{n}\sum_{j=0}^{n-i}\sum_{k=0}^{n-i-j}$. Due to the blend relationship between the parameters of the basis function, the Jacobian polynomials P in (15) with variable parameters are usually used to form the warp tensor product (basis function) [35]. The PKD polynomials are composed of three orthogonal Jacobian polynomials with mixed weights and orders, so the polynomials show hierarchical characteristics and meet

$$\begin{aligned} &\int_{-1}^{1} P_i^{(\alpha,\beta)}(x) P_j^{(\alpha,\beta)}(x)\omega(x)dx \\ &\qquad = \frac{\delta_{ij}2^{\alpha+\beta+1}}{2n+\alpha+\beta+1}\frac{\Gamma(n+\alpha+1)\Gamma(n+\beta+1)}{n!\Gamma(n+\alpha+\beta+1)} \end{aligned} \tag{16}$$

where the weight is $\omega(x) = (1-x)^\alpha(1+x)^\beta$. The Jacobian polynomials of any order can be obtained by recurrence relations.

The PKD polynomials in (15) are defined in a right-angled tetrahedral reference domain $\Omega_s = \{(\xi, \eta, \zeta) | -1 \leq \xi, \eta, \zeta, \quad \xi + \eta + \zeta \leq -1\}$. To establish the relationship between the physical domain $\Omega_e = \{(x, y, z)\}$ and the reference domain, we need to use a smooth and reversible affine transformation [37], as shown in Fig. 1. We assume that the tetrahedral vertices in the physical domain are A(x_1, y_1, z_1) B(x_2, y_2, z_2) C(x_3, y_3, z_3) D(x_4, y_4, z_4). The affine transformation is established through the barycentric coordinates (L_1, L_2, L_3, L_4), which ensures that the relative position between nodes in different coordinate systems remains unchanged. Then, any point Q inside the element

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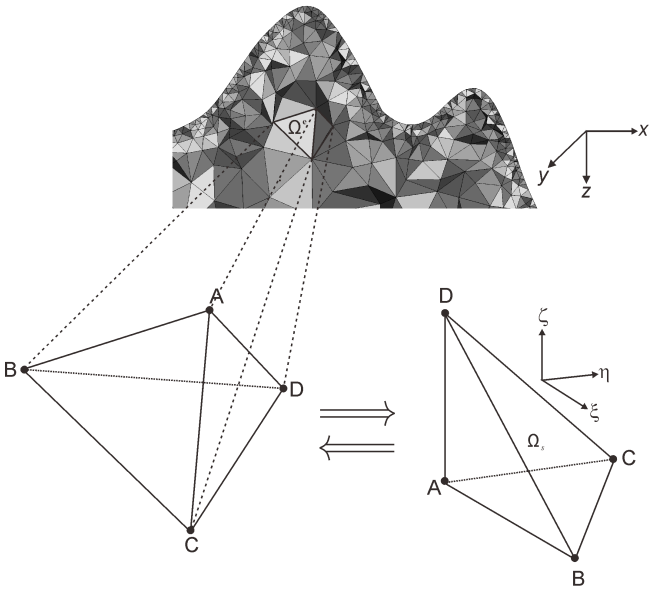


Fig. 1. Affine transformation between physical and reference domains.

connects with four vertices to form the volumes v_1, v_2, v_3, v_4, and its ratio to the whole element v is called the Barycentric coordinates, i.e.,

$$L_1 = \frac{v_1}{v} = \frac{1}{6v}\begin{vmatrix} x & y & z & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$$

$$L_2 = \frac{v_2}{v} = \frac{1}{6v}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x & y & z & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$$

$$L_3 = \frac{v_3}{v} = \frac{1}{6v}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x & y & z & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$$

$$L_4 = \frac{v_4}{v} = \frac{1}{6v}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x & y & z & 1 \end{vmatrix}. \tag{17}$$

As shown in Fig. 2, the points inside the element always satisfy $L_1 + L_2 + L_3 + L_4 = 1, \quad 0 \le L_i \le 1$. Thus, the coordinate of an internal point $\mathrm{Q}(\xi, \eta, \zeta)$ of the element in the reference domain can be expressed as

$$\begin{cases} \xi = -L_3 + L_4 - L_2 - L_1 \\ \eta = -L_3 - L_4 + L_2 - L_1 \\ \zeta = -L_3 - L_4 - L_2 + L_1. \end{cases} \tag{18}$$

Equations (17) and (18) constitute an affine transformation from the physical to reference coordinate system.

The PKD polynomials are multivariate compound functions; the chain rule needs to be used when calculating partial derivatives. For the convenience of calculation, we use collapse transformation to transform the parameters in the

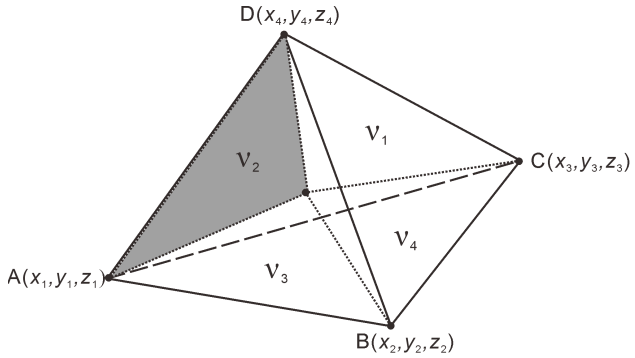


Fig. 2. Barycentric coordinates.

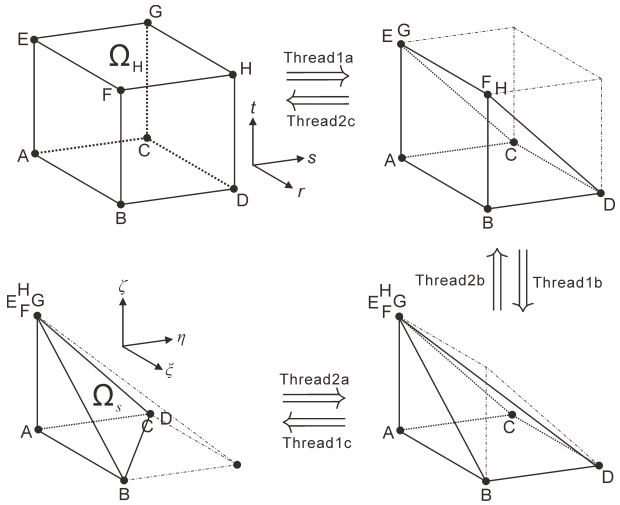


Fig. 3. Collapse transformation (refer to Karniadakis and Sherwin [38]). Thread 1 denotes the transform from the orthogonal coordinate system to reference one, while Thread 2 denotes the inverse transform.

reference domain to the orthogonal hexahedron domain $\Omega_H = \{(r, s, t)| -1 \le r, s, t \le 1\}$.

Referring to Fig. 3, to realize the collapse transform, we first shrink the upper top surface of the hexahedron domain to one edge; accordingly, the hexahedron becomes a prism. Then, we shrink the top edges into a vertex to form a pyramid; finally, the lower bottom quadrilateral of the pyramid is shrunk to a triangle to form a right-angled tetrahedron for the reference domain. The inverse transformation follows the opposite procedure. From the above discussion, we can obtain the following formula for collapse transformation, i.e.:

$$r = -2\frac{1+\xi}{\eta+\zeta} - 1, \quad s = 2\frac{1+\eta}{1-\zeta} - 1, \ t = \zeta \tag{19}$$

and the PKD polynomial can be expressed as

$$\psi_m(\xi, \eta, \zeta) = \sqrt{\frac{8}{\gamma_{ijk}}} P_i^{(0,0)}(r) P_j^{(2i+1,0)}(s)(1-s)^i \times P_k^{(2i+2j+2,0)}(t)(1-t)^{i+j}. \tag{20}$$

This coordinate transformation ensures the tensor property of the basis functions and, thus, is beneficial to numerical integration.

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The PKD polynomials for tetrahedral grids do not have corresponding interpolation nodes, so they need to be constructed separately. If the interpolation nodes in the hexahedral grid are directly mapped into the tetrahedron, the singularity of the basis function at the corner points will cause a large condition number of the equations system and difficulty to solve them [39]. At present, there are not a universally optimal set of interpolation nodes for TSEM (such as the GLL nodes for structured grids). Considering that the research in this article focuses on the numerical simulation of AEM in the frequency domain, what we expect is the good properties and high numerical accuracy of the interpolation nodes, while there is no special requirement on the consistency of the interpolation nodes and the integration ones. Thus, we, respectively, choose the Warp & Blend interpolation nodes and Gaussian integration nodes for our numerical modeling. Among them, the Warp & Blend interpolation nodes adopt an explicit construction format, while the integration nodes adopt a symmetrical cubature point set [37], [40].

In addition, in our TSEM modeling, each node has four degrees of freedom (DOFs), including three components of the vector potential $\mathbf{A}$ and the scalar potential Φ. The distribution of DOFs based on potential equations in a physical element is shown in Fig. 4.

C. Galerkin Weighted Residual Method

To do AEM modeling by TSEM, we first discretize the calculation domain into Ne nonoverlapping tetrahedral elements. Then, we apply the Galerkin weighted residual method to (12) by assuming the weight function to be the basis functions in (14) and obtain

$$\begin{cases} (\nabla N^T, \nabla A_{sx})_\Omega + i\omega\mu_0\sigma(N^T, A_{sx})_\Omega \\ \quad +\mu_0\sigma\left(N^T, \dfrac{\partial \Phi_s}{\partial x}\right)_\Omega \\ = \mu_0\sigma_s(N^T, E_{px})_\Omega \\ (\nabla N^T, \nabla A_{sy})_\Omega + i\omega\mu_0\sigma(N^T, A_{sy})_\Omega \\ \quad +\mu_0\sigma\left(N^T, \dfrac{\partial \Phi_s}{\partial y}\right)_\Omega \\ = \mu_0\sigma_s(N^T, E_{py})_\Omega \\ (\nabla N^T, \nabla A_{sz})_\Omega + i\omega\mu_0\sigma(N^T, A_{sz})_\Omega \\ \quad +\mu_0\sigma\left(N^T, \dfrac{\partial \Phi_s}{\partial z}\right)_\Omega = \mu_0\sigma_s(N^T, E_{pz})_\Omega \\ i\omega\mu_0\sigma(\nabla N^T, \mathbf{A}_s)_\Omega + \mu_0\sigma(\nabla N^T, \nabla\Phi_s)_\Omega \\ \quad = \mu_0\sigma_s(\nabla N^T, \mathbf{E}_p)_\Omega. \end{cases} \tag{21}$$

Since the shape function in each element Ω^e in the calculation domain Ω adopts the high-order spectral element basis functions, we have

$$A^e_{sx} = \sum_{i=1}^{Nt} A^e_{sxi}N_i, \quad A^e_{sy} = \sum_{i=1}^{Nt} A^e_{syi}N_i$$
$$A^e_{sz} = \sum_{i=1}^{Nt} A^e_{szi}N_i, \quad \Phi^e_s = \sum_{i=1}^{Nt} \Phi^e_{si}N_i. \tag{22}$$

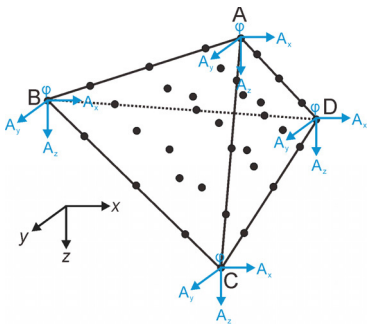


Fig. 4. Fourth order of interpolation nodes and DOFs in a physical element.

Then, we can get the linear equations system for each element by inserting (22) into (21) and doing integrations, i.e.,

$$\mathbf{K}^e\mathbf{u}^e = \mathbf{B}^e \tag{23}$$

where $\mathbf{u}^e = (A^e_{sx}, A^e_{sy}, A^e_{sz}, \Phi^e_s)^T$ are the potentials to be solved. The symmetric matrix $\mathbf{K}^e$ and the source term can be expressed as

$$\mathbf{K}^e = \begin{bmatrix} K^e_{11} & 0 & 0 & K^e_{14} \\ 0 & K^e_{22} & 0 & K^e_{24} \\ 0 & 0 & K^e_{33} & K^e_{34} \\ K^e_{41} & K^e_{42} & K^e_{43} & K^e_{44} \end{bmatrix} \tag{24}$$

$$\mathbf{B}^e = \left(b^e_1,\ b^e_2,\ b^e_3,\ b^e_4\right)^T. \tag{25}$$

The expressions of the subitems in (24) and (25) are given in the Appendix. Assembling (23) to all elements, the global matrix equation for the potentials can be written as

$$\mathbf{K}\mathbf{u} = \mathbf{B}. \tag{26}$$

Since AEM systems are multisource, meaning that the transmitters and receivers move synchronously, we use the multifrontal direct solver MUMPS [41] to solve the linear equation (26) for the vector potential $\mathbf{A}$ and scalar potential Φ at each node and then calculate the EM field by using (2) and (3).

III. Numerical Experiments

A. Accuracy Verification

In order to check the accuracy of our TSEM for AEM modeling, we design a three-layer model in Fig. 5(a). The first layer has a resistivity of $\rho_1 = 100\ \Omega\cdot$m and a thickness of 30 m; the second layer has a resistivity of $\rho_2 = 1\ \Omega\cdot$m and a thickness of 30 m; and the basement has a resistivity of $\rho_3 = 100\ \Omega\cdot$m. We take the horizontal coplanar (HCP) array as example in our modeling. The Tx–Rx offset is 10 m, the flight altitude of the coil system is 30 m, and the transmitting dipole moment is 1 Am^2. We use the third-order ($n = 3$) TSEM to calculate EM responses at 19 frequencies from 100 Hz to 100 kHz. Fig. 5(b) and (c) shows the comparison between the results from this article and the semianalytical solutions for a layered Earth. It is seen that our results match well the semianalytical solutions, with the relative errors of both real and imaginary parts less than 1%.

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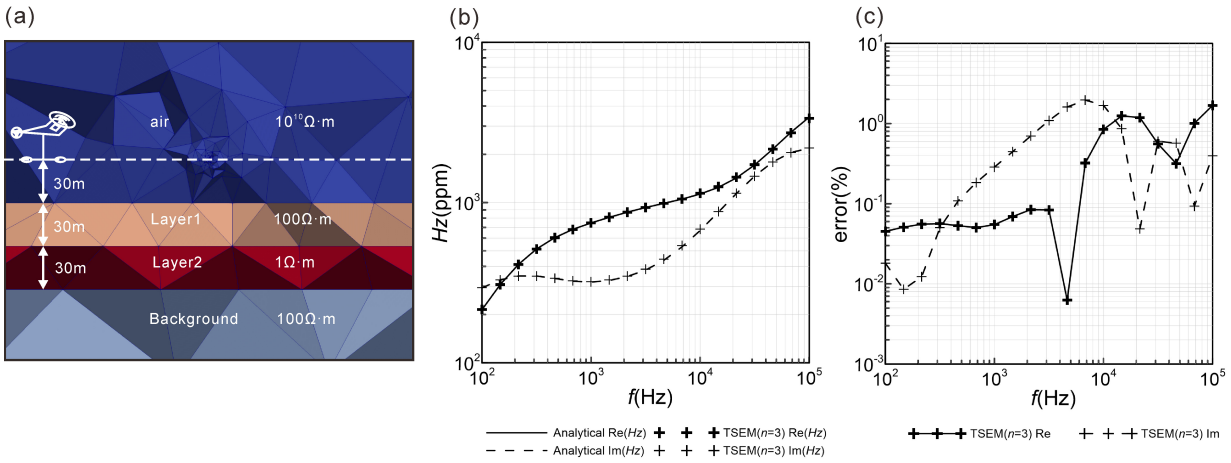


Fig. 5. Comparison between our results and the semianalytical solutions for a layered Earth model. (a) model and grids. (b) Real and imaginary parts. (b) Relative error.

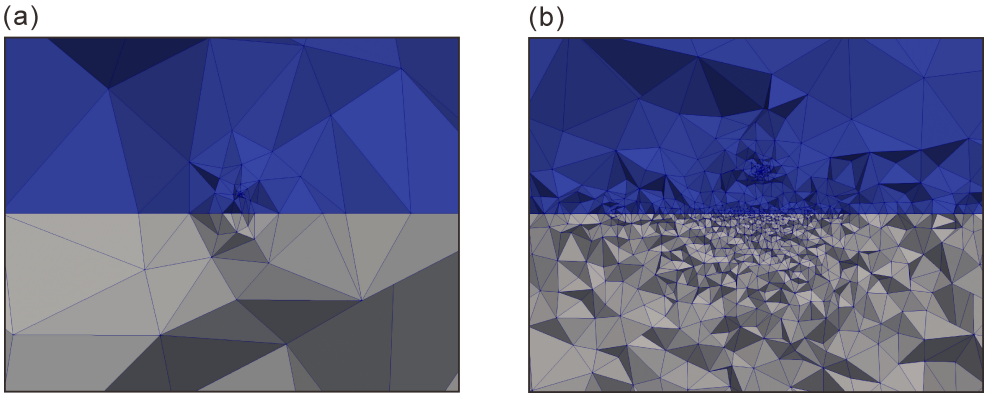


Fig. 6. Homogeneous half-space model. (a) Initial grids. (b) Grids after seven times adaptive refinements.

B. Efficiency Analysis

In the following, we will analyze the efficiency of our TSEM using a half-space model of 100 $\Omega \cdot$ m. We use again the HCP coil array with the following parameters: the transmitter is located at (0, 0, −30 m) with a dipole moment of 1 Am^2, and the Tx–Rx offset is 10 m. For a single source, the EM responses for 19 frequencies are calculated, respectively, by the goal-oriented adaptive FE method (AFEM) and our TSEM ($n = 1$, $n = 2$, $n = 3$, and $n = 4$). The adaptive grid refinement adopts the strategy proposed by Yin *et al.* [42]. When the minimum volume constraint is reached and the weighted posterior error is less than a threshold, the grid refinement will be terminated. For the present model, we update the grids seven times. The first set of (original) grids and the eighth set of refined grids are shown in Fig. 6. It is seen that, after refinement, the grids near the receiver and close to the surface below the receiver become very dense.

Based on the unstructured grids in Fig. 6, we, respectively, use our TSEM and AFEM to calculate the EM responses. The comparison of the results is shown in Fig. 7. Among them, Fig. 7(a) and (d) shows the relative errors between the 1-D semianalytical solutions and those from nodal AFEM based on EM potentials, and Fig. 7(b) and (e) shows the relative errors between the semianalytical solutions and edge AFEM based on EM fields, while Fig. 7(c) and (f) shows the relative errors between the semianalytical solutions and TSEM based on EM potentials. It is seen that both grid refinement and increasing TSEM order can improve the accuracy of numerical simulation. For both nodal and edge elements, the modeling accuracy depends heavily on grids. For the coarse grid, neither of them can achieve accurate results. However, using TSEM of third order ($n = 3$), we can obtain high computational accuracy even for the coarsest first set of grids.

Tables I and II show the parameter comparison of TSEM and nodal AFEM and edge AFEM (c.f. Fig. 7). The edge AFEM adopts vector basis functions and locates unknowns at the edges, while the nodal AFEM and TSEM from this article solve the potentials at the nodes. The initial grids of the three algorithms are the same. From the number of grids, nodes, DOFs, the average and maximum relative errors for real and imaginary parts, the running time (counted for solving the linear systems by the PARDISO solver), and the maximum memory cost for each frequency, as shown in Tables I and II, we draw the following conclusions.

1) With the grid refinement, the modeling accuracies of both nodal and edge AFEM are continuously improved, and the accuracy of edge AFEM is higher than the nodal AFEM, while the time and memory requirements are less than that of the nodal AFEM. This is because,

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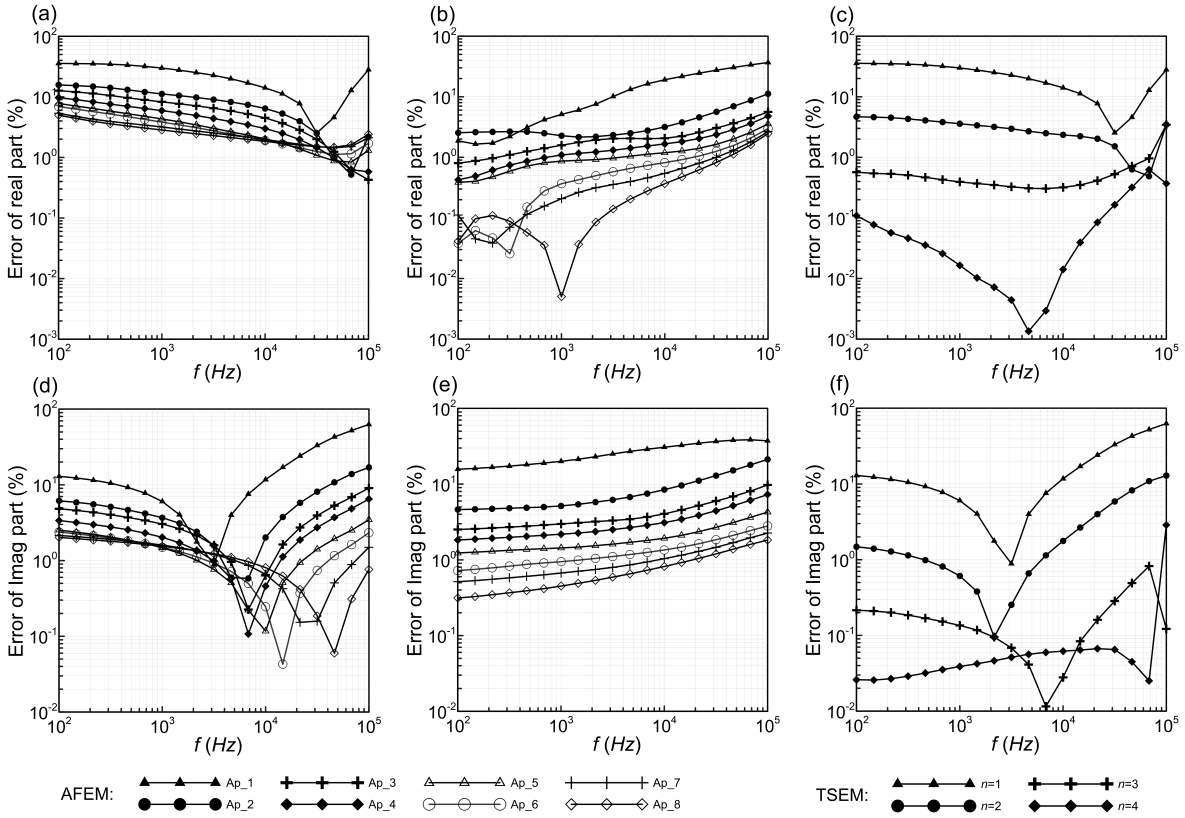


Fig. 7. Relative error between semianalytical solutions and unstructured AFEM and our TSEM. (a) and (d) Results for different grid refinements using nodal AFEM based on potentials. (b) and (e) Results for different grid refinements using edge AFEM based on EM fields. (c) and (f) Results for different orders of TSEM based on potentials.

TABLE I
COMPARISON BETWEEN NODAL AFEM AND TSEM

Algorithm	AFEM (Nodal element)								TSEM			
	Ap_1	Ap_2	Ap_3	Ap_4	Ap_5	Ap_6	Ap_7	Ap_8	n=1	n=2	n=3	n=4
Number of Grids	2096	8287	15133	25840	48334	92325	114169	172201	2096	2096	2096	2096
Number of Nodes	386	1393	2495	4197	7773	12728	18181	27421	386	2925	9714	22849
DOFs	1544	5572	9980	16788	31092	50912	72724	109684	1544	11700	38856	91396
Average relative error (%) (Re/Imag)	22.60 17.51	8.57 5.52	6.37 3.46	4.58 2.38	3.47 1.52	3.21 1.31	2.80 1.23	2.62 1.15	22.60 17.51	3.00 2.98	0.62 0.19	0.11 0.19
Maximum relative error (%) (Re/Imag)	35.61 62.93	15.72 17.01	12.64 9.06	9.73 6.50	7.76 3.46	6.93 2.39	5.34 2.15	4.97 2.02	35.61 62.93	4.67 12.93	3.44 0.82	0.62 2.87
Running time (s)	0.024	0.15	0.29	0.51	1.32	2.59	5.37	10.95	0.029	0.40	2.09	9.20
Maximum memory (MB)	24	50	137	225	479	837	1374	2220	26	183	808	2640

in one element, there are six DOFs for the edge AFEM and 16 DOFs for the nodal AFEM. Therefore, the number of relevant DOFs to each unknown in the nodal AFEM is larger than that in the edge AFEM. This results in a denser coefficient matrix that increases the running time and memory consumption. Similarly, with increasing order, the running time and memory cost for TSEM are bigger than the nodal AFEM for the same grids. This is again caused by the substantial growth of relevant DOFs.

2) The comparison of Ap_6 of nodal AFEM with TSEM ($n = 3$) shows that, with similar running time and

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TABLE II
COMPARISON BETWEEN EDGE AFEM AND TSEM

Algorithm	AFEM (Edge element)								TSEM			
	Ap_1	Ap_2	Ap_3	Ap_4	Ap_5	Ap_6	Ap_7	Ap_8	n=1	n=2	n=3	n=4
Number of Grids	2096	8287	15133	25840	48334	92325	114169	172201	2096	2096	2096	2096
Number of Nodes	386	1393	2495	4197	7773	12728	18181	27421	386	2925	9714	22849
DOFs	2539	9774	17727	30139	56224	79478	132471	199744	1544	11700	38856	91396
Average relative error (%) (Re/Imag)	14.04 26.16	3.89 8.53	2.18 4.23	1.63 3.16	1.20 1.98	0.77 1.32	0.59 1.00	0.46 0.77	22.60 17.51	3.00 2.98	0.62 0.19	0.11 0.19
Maximum relative error (%) (Re/Imag)	36.70 38.60	11.21 21.23	5.61 9.75	4.80 7.28	3.57 4.28	2.97 2.80	2.66 2.26	2.46 1.82	35.61 62.93	4.67 12.93	3.44 0.82	0.62 2.87
Running time (s)	0.024	0.14	0.29	0.52	1.23	2.27	4.32	7.73	0.029	0.40	2.09	9.20
Maximum memory (MB)	24	47	125	218	430	758	1253	1954	26	183	808	2640

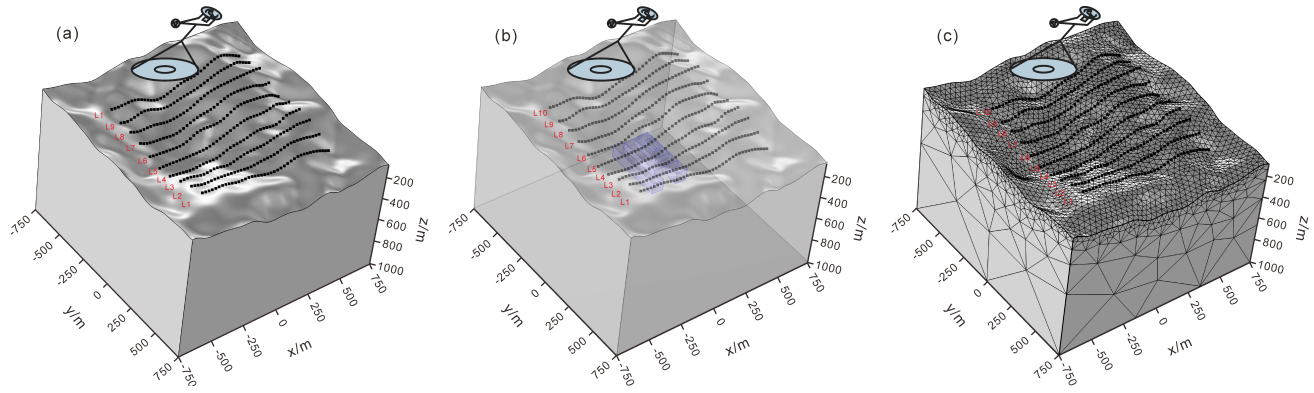


Fig. 8. Topographic Earth with an A-shaped abnormal body embedded. (a) Topography and survey line. (b) Location of abnormal body. (c) Grid.

memory cost, the accuracy of the real part from TSEM is about five times that of AFEM, while, for the imaginary part, it is about seven times. Further comparison of Ap_8 of nodal AFEM and TSEM ($n = 4$) shows that the TSEM takes less time, but the accuracy of the real part of TSEM is more than 20 times of AFEM and six times for the imaginary part. Compared with the grid refinement, increasing TSEM order has greater advantages in improving the accuracy.

3) When the edge AFEM is used to refine the mesh, the convergence rate gradually slows down. Even if the running time and memory cost are greatly increased, the accuracy cannot be improved remarkably. In contrast, the TSEM can still increase accuracy significantly when the order is increased. In particular, the running time for TSEM ($n = 3$) is a little less than that for the edge AFEM (Ap_6), but the relative error of the real part calculated by TSEM ($n = 3$) is 0.81 times that of edge AFEM (Ap_6), while, for the imaginary part, it is only 1/7. This implies that, using TSEM, we can obtain higher accuracy with less computational cost. Besides, we notice from the tables that the maximum relative error of the imaginary part in the TSEM fourth order is bigger than that of the third order. This can also be seen from Fig. 7 around 10^4 or at 10^5 Hz. The reason for this may be that, at some points, the EM field does not change so sharply; thus, using high-order interpolation functions may result in overfitting.

From the above discussions, we conclude that the TSEM algorithm proposed in this article has obvious advantages in an accuracy improvement. In practical forward modeling, we need to coordinate between modeling efficiency and accuracy. This means that we can choose the appropriate orders for TSEM according to the accuracy requirements so that the efficiency can be improved while ensuring calculation accuracy.

C. Complex Geological Model

The tetrahedral grids have good adaptability to topography and complex underground structures. To demonstrate this, we design a model in Fig. 8 with an A-shaped anomalous body embedded under a topographic Earth surface. We set ten survey lines at a line spacing of 100 m. Each survey line has

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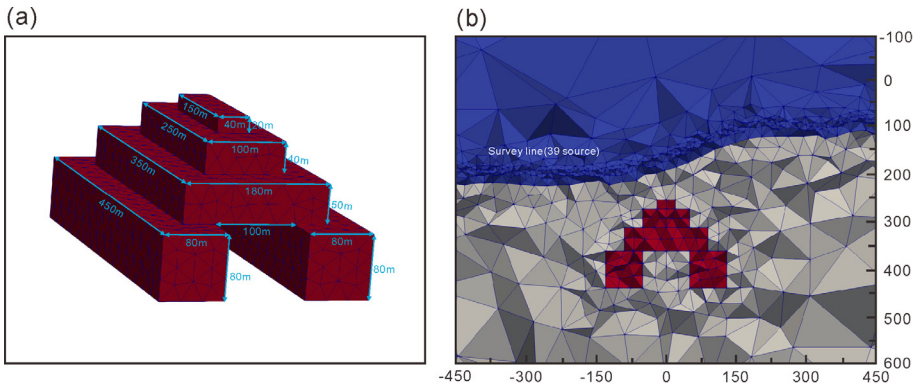


Fig. 9. Abnormal body model. (a) Abnormal body size and grid. (b) Grids corresponding to the sixth survey line.

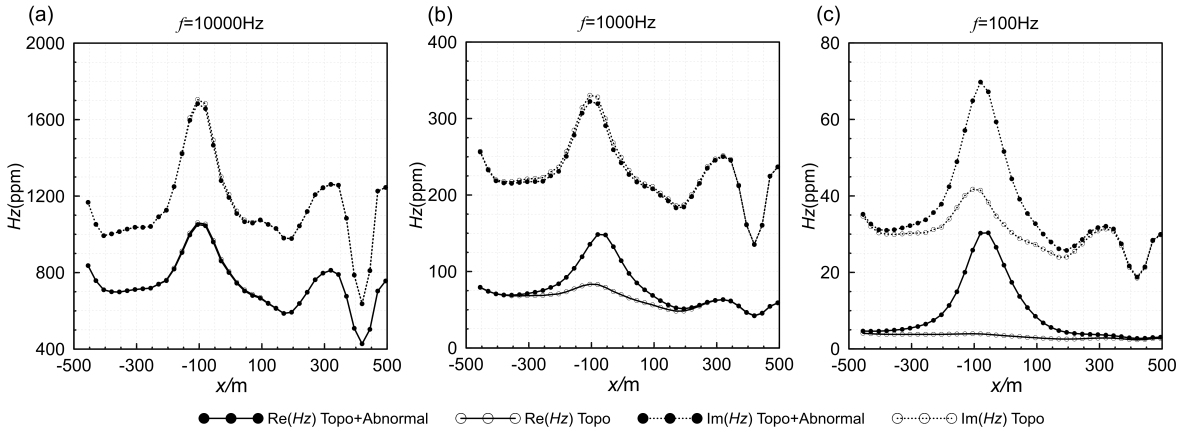


Fig. 10. Comparison of EM responses for purely topographic Earth and topographic Earth with an embedded abnormal body. The frequencies are (a) 10 000, (b) 1000, and (c) 100 Hz.

39 points at an interval of 25 m. We adopt an HCP coil array for our modeling. The flight height of the system along the survey line is always kept at 30 m, the Tx–Rx offset is 10 m, and the frequencies are set to be 100, 1000, and 10 000 Hz. The half-space has a resistivity of 100 $\Omega \cdot$ m, while the abnormal body has a resistivity of 1 $\Omega \cdot$m. The abnormal body is located at the center of the survey line and is symmetrical about $x = 0$. Refer to Fig. 9 for the size and gridding of the abnormal body. We consider, respectively, the cases of pure topography and topography with abnormal body embedded.

Taking the sixth survey line as an example, we study the influence of topography and abnormal body on AEM responses. The section along the survey line is shown in Fig. 9(b). The EM responses for a purely topographic Earth and topographic Earth with an embedded abnormal body are shown in Fig. 10. It is seen that, at high frequency $f = 10\,000$ Hz, little influence from the abnormal body can be seen, and the EM responses for two cases with and without an abnormal body are virtually not distinguishable. This is due to the fact that the high-frequency EM wave can only penetrate shallowly. Thus, the EM field at this time has not yet penetrated into the abnormal body, so the EM signal mainly comes from the conductive background. With decreasing frequency, the EM signal penetrates into the deep Earth, and the influence of the abnormal body gradually increases, so we can see the difference of EM responses in Fig. 10(b). When the frequency is reduced to $f = 100$ Hz, the influence of the abnormal body dominates, and the response difference between the two models becomes quite large. Moreover, we can see from the figures that, due to the undulating Earth surface, the peak position of EM responses does not coincide with the center of the abnormal body but, instead, moves toward the topographic inclination.

To intuitively illustrate the influence of the topography on AEM responses, we also show in Figs. 11 and 12 the contour maps of multisource and multifrequency AEM responses for purely topographic Earth and topographic Earth with an embedded anomaly body. Comparing the two figures, we can draw the same conclusions as in Fig. 10. At high frequencies, the distribution of the EM fields is essentially not affected by the abnormal body, but the topographic effect dominates. As the frequency decreases, the influence of the conductive abnormal body becomes obvious. Especially, the real part of EM responses shows a peak anomaly consistent with the abnormal body. As the frequency further decreases, the EM field penetrates into the deep Earth, and the EM responses are mainly contributed by the abnormal body. The real and imaginary parts of the responses (especially the real part) show clear anomalies from the abnormal body.

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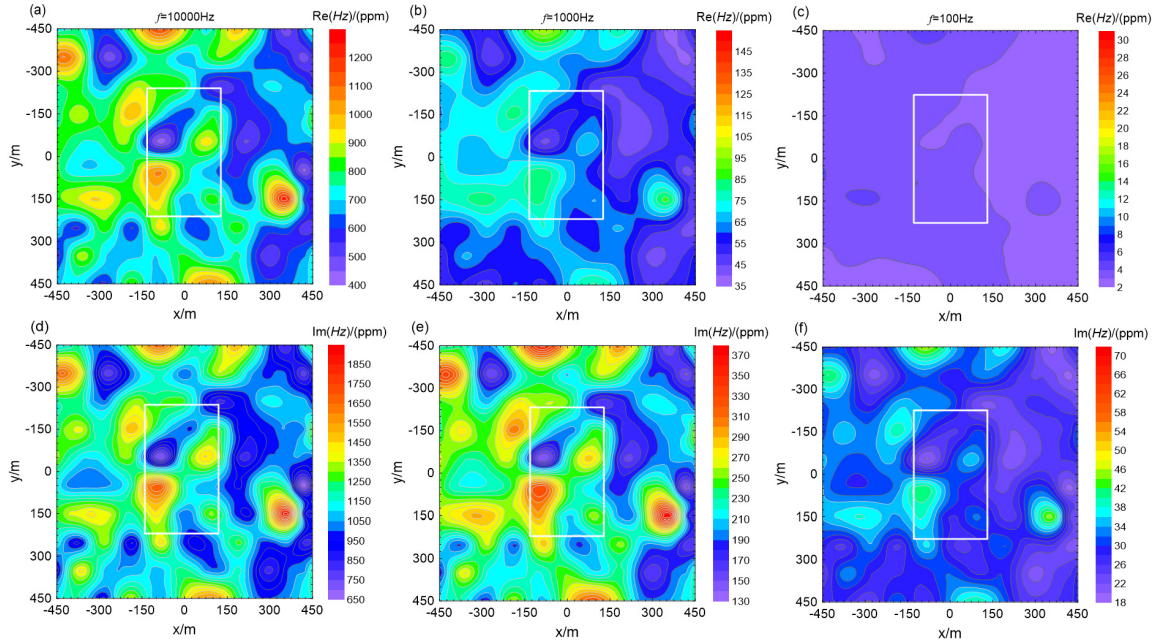


Fig. 11. AEM responses for the purely topographic Earth in Fig.8 (the white box indicates the position of the abnormal body). (a)–(c) Real parts for frequencies of 10 000, 1000, and 100 Hz, respectively. (d)–(f) Imaginary parts for the same frequencies.

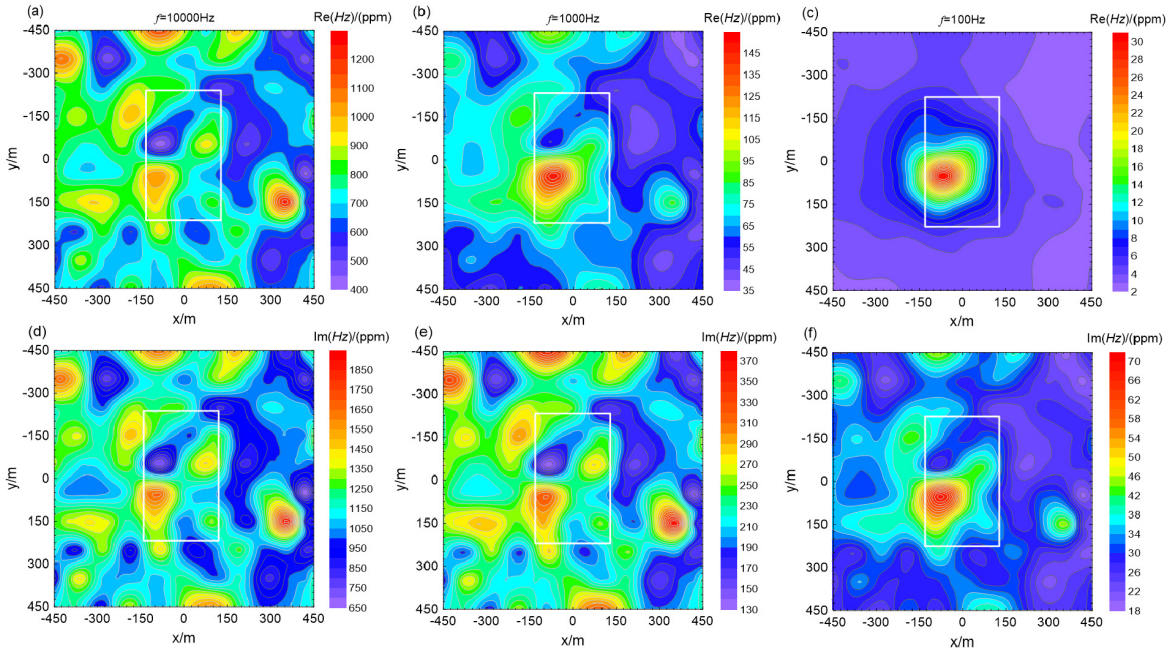


Fig. 12. AEM responses for the topographic Earth with an embedded abnormal body in Fig. 9. (a)–(c) Real parts for frequencies of 10 000, 1000, and 100 Hz. (d)–(f) Imaginary parts for the same frequencies.

IV. Conclusion

In this article, we have successfully developed a forward modeling algorithm based on EM potentials and the unstructured nodal TSEM, and applied it to the simulation of AEM systems in the frequency domain. Via a layered Earth model, we have validated the accuracy of the algorithm. In contrast to the conventional AFEM in which the modeling accuracy is heavily dependent of the grids, the TSEM can improve the numerical accuracy by changing the order of the basis functions without depending much on the grids. By choosing appropriate orders, our TSEM can obtain high modeling accuracy even on coarse grids. The numerical results for a topographic Earth with abnormal body embedded showed that the EM responses are seriously affected by the topography, especially at high frequencies. To detect the target in the deep Earth, attentions need to be paid to the low-frequency signal. It is worth noting that the SEM discussed in this article is based on the EM potential of the nodal element. It can be

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imagined that the direct solution of EM field based on edge element will surely have higher accuracy, which will be our future research focus.

APPENDIX

It can be seen from the previous discussion that, when using TSEM to simulate AEM responses, we need to calculate the elements of the coefficient matrix and the source term in (24) and (25). For this purpose, we take (20)–(22) and obtain

$$K_{11}^e = K_{22}^e = K_{33}^e = \iiint_{\Omega}\left[\left(\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}+\frac{\partial N_i}{\partial y}\frac{\partial N_j}{\partial y}+\frac{\partial N_i}{\partial z}\frac{\partial N_j}{\partial z}\right)+i\omega\mu_0\sigma N_i N_j\right]d\Omega \tag{27}$$

$$K_{14}^e = \mu_0\sigma\iiint_{\Omega} N_i\frac{\partial N_j}{\partial x}d\Omega \tag{28}$$

$$K_{24}^e = \mu_0\sigma\iiint_{\Omega} N_i\frac{\partial N_j}{\partial y}d\Omega \tag{29}$$

$$K_{34}^e = \mu_0\sigma\iiint_{\Omega} N_i\frac{\partial N_j}{\partial z}d\Omega \tag{30}$$

$$K_{41}^e = i\omega\mu_0\sigma\iiint_{\Omega}\frac{\partial N_i}{\partial x}N_j d\Omega \tag{31}$$

$$K_{42}^e = i\omega\mu_0\sigma\iiint_{\Omega}\frac{\partial N_i}{\partial y}N_j d\Omega \tag{32}$$

$$K_{43}^e = i\omega\mu_0\sigma\iiint_{\Omega}\frac{\partial N_i}{\partial z}N_j d\Omega \tag{33}$$

$$K_{44}^e = \mu_0\sigma\iiint_{\Omega}\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}+\frac{\partial N_i}{\partial y}\frac{\partial N_j}{\partial y}+\frac{\partial N_i}{\partial z}\frac{\partial N_j}{\partial z}d\Omega \tag{34}$$

$$b_1^e = \mu_0\sigma_s\sum_{j=1}^{Nt}E_{pxj}\iiint_{\Omega} N_i N_j d\Omega \tag{35}$$

$$b_2^e = \mu_0\sigma_s\sum_{j=1}^{Nt}E_{pyj}\iiint_{\Omega} N_i N_j d\Omega \tag{36}$$

$$b_3^e = \mu_0\sigma_s\sum_{j=1}^{Nt}E_{pzj}\iiint_{\Omega} N_i N_j d\Omega \tag{37}$$

$$b_4^e = \mu_0\sigma_s\sum_{j=1}^{Nt}\left(E_{pxj}\iiint_{\Omega}\frac{\partial N_i}{\partial x}N_j d\Omega + E_{pyj}\iiint_{\Omega}\frac{\partial N_i}{\partial y}N_j d\Omega + E_{pzj}\iiint_{\Omega}\frac{\partial N_i}{\partial z}N_j d\Omega\right). \tag{38}$$

In the above equations, the integral of PKD polynomials has no analytical solution, so we need to use Gaussian numerical integration. The integrals in (27) and (38) can be divided into three types that are expressed as K_{ij}^1, K_{ij}^2, and K_{ij}^3, that is,

$$\begin{aligned}K_{ij}^1 &= \iiint_{\Omega_e}\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}\mathrm{d}\Omega = \iiint_{\Omega_e}\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}|\mathbf{J}_{3\mathrm{D}}|\mathrm{d}\xi\mathrm{d}\eta\mathrm{d}\zeta\\ &\approx |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\left(\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}\right)_m\\ &= |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\\ &\quad\times\left(\left[\frac{\partial\psi_k(\xi_m,\eta_m,\zeta_m)}{\partial\xi}\frac{\partial\xi}{\partial x}+\frac{\partial\psi_k(\xi_m,\eta_m,\zeta_m)}{\partial\eta}\frac{\partial\eta}{\partial x}+\frac{\partial\psi_k(\xi_m,\eta_m,\zeta_m)}{\partial\zeta}\frac{\partial\zeta}{\partial x}\right]\right)\\ &\quad\times\left[\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\xi}\frac{\partial\xi}{\partial x}+\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\eta}\frac{\partial\eta}{\partial x}+\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\zeta}\frac{\partial\zeta}{\partial x}\right]\end{aligned} \tag{39}$$

$$\begin{aligned}K_{ij}^2 &= \iiint_{\Omega_e}N_i\frac{\partial N_j}{\partial x}\mathrm{d}\Omega = \iiint_{\Omega_e}N_i\frac{\partial N_j}{\partial x}|\mathbf{J}_{3\mathrm{D}}|\mathrm{d}\xi\mathrm{d}\eta\mathrm{d}\zeta\\ &\approx |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\left(N_i\frac{\partial N_j}{\partial x}\right)_m\\ &= |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\\ &\quad\times\left(\psi_k(\xi_m,\eta_m,\zeta_m)\times\left[\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\xi}\frac{\partial\xi}{\partial x}+\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\eta}\frac{\partial\eta}{\partial x}+\frac{\partial\psi_l(\xi_m,\eta_m,\zeta_m)}{\partial\zeta}\frac{\partial\zeta}{\partial x}\right]\right)\end{aligned} \tag{40}$$

$$\begin{aligned}K_{ij}^3 &= \iiint_{\Omega_e}N_i N_j\mathrm{d}\Omega = \iiint_{\Omega_e}N_i N_j|\mathbf{J}_{3\mathrm{D}}|\mathrm{d}\xi\mathrm{d}\eta\mathrm{d}\zeta\\ &\approx |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\left(N_i N_j\right)_m\\ &= |\mathbf{J}_{3\mathrm{D}}|\sum_{k,l=1}^{Nt}(\mathbf{V}^{-1})_{ki}(\mathbf{V}^{-1})_{lj}\sum_{m=1}^{Nf}\omega_m\\ &\quad\times\psi_k(\xi_m,\eta_m,\zeta_m)\times\psi_l(\xi_m,\eta_m,\zeta_m)\end{aligned} \tag{41}$$

where *Nf* denotes the number of nodes in numerical Gaussian integration, w_i are the weights, and $\mathbf{J}_{3\mathrm{D}}$ is the Jacobian matrix for the transformation from the physical to the reference coordinate system.

REFERENCES

[1] A. Viezzoli, L. Tosi, P. Teatini, and S. Silvestri, "Surface water–groundwater exchange in transitional coastal environments by airborne electromagnetics: The Venice Lagoon example," *Geophys. Res. Lett.*, vol. 37, no. 1, Jan. 2010, Art. no. L01402.

[2] C. G. Farquharson and D. W. Oldenburg, "An integral equation solution to the geophysical electromagnetic forward-modelling problem," in *Methods in Geochemistry and Geophysics*, vol. 35. Amsterdam, The Netherlands: Elsevier, Nov. 2002, pp. 3–19.

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[3] G. A. Newman and D. L. Alumbaugh, "Three-dimensional induction logging problems. Part 2: A finite-difference solution," *Geophysics*, vol. 67, no. 2, pp. 484–491, Mar. 2002.
[4] C. J. Weiss and S. Constable, "Mapping thin resistors and hydrocarbons with marine EM methods. Part II—Modeling and analysis in 3D," *Geophysics*, vol. 71, no. 6, pp. G321–G332, Nov. 2006.
[5] H. C. Jahandari and G. Farquharson, "Finite-volume modelling of geophysical electromagnetic data on unstructured grids using potentials," *Geophys. J. Int.*, vol. 202, no. 3, pp. 1859–1876, Sep. 2015.
[6] R. D. Graglia and I.-L. Gheorma, "Higher order interpolatory vector bases on pyramidal elements," *IEEE Trans. Antennas Propag.*, vol. 47, no. 5, pp. 775–782, May 1999.
[7] J. M. Jin, *The Finite Element Method in Electromagnetics*, 2nd ed. Hoboken, NJ, USA: Wiley, 2002.
[8] J. C. Nédélec, "Mixed finite element in $\mathbb{R}^3$," *Numerische Math.*, vol. 35, pp. 315–341, Sep. 1980.
[9] T. Kang, T. Chen, Y. Wang, and K. I. Kim, "A T-ψ formulation with the penalty function term for the 3D eddy current problem in laminated structures," *Appl. Math. Comput.*, vol. 271, pp. 618–641, Nov. 2015.
[10] E. A. Badea, M. E. Everett, G. A. Newman, and O. Biro, "Finite-element analysis of controlled-source electromagnetic induction using Coulomb-gauged potentials," *Geophysics*, vol. 66, no. 3, pp. 786–799, Jan. 2001.
[11] V. Puzyrev, J. Koldan, J. de la Puente, G. Houzeaux, M. Vázquez, and J. M. Cela, "A parallel finite-element method for three-dimensional controlled-source electromagnetic forward modelling," *Geophys. J. Int.*, vol. 193, no. 2, pp. 678–693, Apr. 2013.
[12] S. M. Ansari, C. G. Farquharson, and S. P. MacLachlan, "A gauged finite-element potential formulation for accurate inductive and galvanic modelling of 3-D electromagnetic problems," *Geophys. J. Int.*, vol. 210, no. 1, pp. 105–129, Jul. 2017.
[13] E. Haber, U. M. Ascher, D. A. Aruliah, and D. W. Oldenburg, "Fast simulation of 3D electromagnetic problems using potentials," *J. Comput. Phys.*, vol. 163, no. 1, pp. 150–171, Sep. 2000.
[14] J. D. Jackson, "From Lorenz to Coulomb and other explicit gauge transformations," *Amer. J. Phys.*, vol. 70, no. 9, pp. 917–928, Sep. 2002.
[15] J. Long and C. G. Farquharson, "On the forward modelling of three-dimensional magnetotelluric data using a radial-basis-function-based mesh-free method," *Geophys. J. Int.*, vol. 219, no. 1, pp. 394–416, Jul. 2019.
[16] J. Zhu *et al.*, "3-D DC resistivity modelling based on spectral element method with unstructured tetrahedral grids," *Geophys. J. Int.*, vol. 220, no. 3, pp. 1748–1761, Mar. 2020.
[17] H. Chen and T. Li, "3-D marine controlled-source electromagnetic modelling in an anisotropic medium using a wavelet–Galerkin method with a secondary potential formulation," *Geophys. J. Int.*, vol. 219, no. 1, pp. 373–393, Oct. 2019.
[18] Z. Ren, L. Qiu, J. Tang, X. Wu, X. Xiao, and Z. Zhou, "3-D direct current resistivity anisotropic modelling by goal-oriented adaptive finite element methods," *Geophys. J. Int.*, vol. 212, no. 1, pp. 76–87, Jan. 2018.
[19] C. Canuto, M. Y. Hussani, A. Quarteroni, and T. A. Zang, *Spectral Methods: Fundamentals in Single Domains*. Berlin, Germany: Springer-Verlag, 1988.
[20] C. Runge, "Über empirische funktionen und die interpolation zwischen äquidistanten ordinaten," *Z. Angew. Math. Phys.*, vol. 46, pp. 224–243, Jan. 1901.
[21] A. T. Patera, "A spectral element method for fluid dynamics: Laminar flow in a channel expansion," *J. Comput. Phys.*, vol. 54, no. 3, pp. 468–488, Jun. 1984.
[22] E. M. Rønquist and A. T. Patera, "A Legendre spectral element method for the Stefan problem," *Int. J. Numer. Methods Eng.*, vol. 24, no. 12, pp. 2273–2299, Dec. 1987.
[23] G. Seriani, "Double-grid Chebyshev spectral elements for acoustic wave modeling," *Wave Motion*, vol. 39, no. 4, pp. 351–360, Apr. 2004.
[24] J.-H. Lee, T. Xiao, and Q. H. Liu, "A 3-D spectral-element method using mixed-order curl conforming vector basis functions for electromagnetic fields," *IEEE Trans. Microw. Theory Techn.*, vol. 54, no. 1, pp. 437–444, Jan. 2006.
[25] Y. Liu, J. Teng, T. Xu, and J. Badal, "Higher-order triangular spectral element method with optimized cubature points for seismic wavefield modeling," *J. Comput. Phys.*, vol. 336, pp. 458–480, May 2017.
[26] D. Komatitsch, C. Barnes, and J. Tromp, "Simulation of anisotropic wave propagation based upon a spectral element method," *Geophysics*, vol. 65, no. 4, pp. 1251–1260, Jul. 2000.
[27] D. Komatitsch and J. Tromp, "Spectral-element simulations of global seismic wave propagation—I. Validation," *Geophys. J. Int.*, vol. 149, no. 2, pp. 390–412, May 2002.
[28] D. Komatitsch and J. Tromp, "A perfectly matched layer absorbing boundary condition for the second-order seismic wave equation," *Geophys. J. Int.*, vol. 154, no. 1, pp. 146–153, Jul. 2003.
[29] Y. Liu, J. Teng, H. Lan, X. Si, and X. Ma, "A comparative study of finite element and spectral element methods in seismic wavefield modeling," *Geophysics*, vol. 79, no. 2, pp. T91–T104, Mar. 2014.
[30] M. Roland *et al.*, "A high-order 3-D spectral-element method for the forward modelling and inversion of gravimetric data—Application to the western Pyrenees," *Geophys. J. Int.*, vol. 209, no. 1, pp. 406–424, Apr. 2017.
[31] C. Yin, X. Huang, Y. Liu, and J. Cai, "3-D modeling for airborne EM using the spectral-element method," *J. Environ. Eng. Geophys.*, vol. 22, no. 1, pp. 13–23, Mar. 2017.
[32] L. Liu, C. Yin, Y. Liu, C. Qiu, and B. Zhang, "Spectral element method for 3D frequency-domain marine controlled-source electromagnetic forward modeling," (in Chinese), *Chin. J. Geophys.*, vol. 61, no. 2, pp. 756–766, Feb. 2018.
[33] X. Huang *et al.*, "Spectral-element method with arbitrary hexahedron meshes for time-domain 3D airborne electromagnetic forward modeling," *Geophysics*, vol. 84, no. 1, pp. E37–E46, Jan./Feb. 2019.
[34] Y. Maday and E. M. Rønquist, "Optimal error analysis of spectral methods with emphasis on non-constant coefficients and deformed geometries," *Comput. Methods Appl. Mech. Eng.*, vol. 80, nos. 1–3, pp. 91–115, Jun. 1990.
[35] J. S. Hesthaven and T. Warburton, "Nodal high-order methods on unstructured grids: I. Time-domain solution of Maxwell's equations," *J. Comput. Phys.*, vol. 181, no. 1, pp. 186–221, Sep. 2002.
[36] M. Dumbser and M. Käser, "An arbitrary high-order discontinuous Galerkin method for elastic waves on unstructured meshes—II. The three-dimensional isotropic case," *Geophys. J. Int.*, vol. 167, no. 1, pp. 319–336, Oct. 2006.
[37] T. Warburton, "An explicit construction of interpolation nodes on the simplex," *J. Eng. Math.*, vol. 56, no. 3, pp. 247–262, Jan. 2007.
[38] G. E. Karniadakis and S. J. Sherwin, *Spectral/hp Element Methods for Computational Fluid Dynamics*. (Numerical Mathematics and Scientific Computation). New York, NY, USA: Oxford Univ. Press, 1999.
[39] W. Heinrichs, "Improved Lebesgue constants on the triangle," *J. Comput. Phys.*, vol. 207, no. 2, pp. 625–638, Aug. 2005.
[40] L. Zhang, T. Cui, and H. Liu, "A set of symmetric quadrature rules on triangles and tetrahedra," *J. Comput. Math.*, vol. 27, no. 1, pp. 89–96, Jan. 2009.
[41] P. R. Amestoy, I. S. Duff, and J.-Y. L'Excellent, "Multifrontal parallel distributed symmetric and unsymmetric solvers," *Comput. Methods Appl. Mech. Eng.*, vol. 184, nos. 2–4, pp. 501–520, Apr. 2000.
[42] C. Yin, B. Zhang, Y. Liu, and J. Cai, "A goal-oriented adaptive finite-element method for 3D scattered airborne electromagnetic method modeling," *Geophysics*, vol. 81, no. 5, pp. E337–E346, Sep. 2016.



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